

Addressing Hawai'i Electricity Misconceptions: Renewable Energy is Expensive, Costs Taxpayers and Ratepayers Significantly, Comes at the Expense of Infrastructure Maintenance, and Leads to Less Reliable Service

Portfolio edition: minor spelling, formatting, and privacy cleanup only. Content reflects the original date and context.

In 2001, Hawai'i adopted one of the most ambitious statewide Renewable Portfolio Standards (RPS), and as updated in 2022, it requires electric utilities to source 100% of their generation from renewable resources by the year 2045. The RPS is further reinforced by the ability of the PUC to impose penalties for underachievement "provided that if the commission determines that the electric utility company is unable to meet the renewable portfolio standards because of reasons beyond the reasonable control of the electric utility company." Common rhetoric around the RPS has followed, citing a familiar story that renewable energy is unreliable, expensive, and increases costs to ratepayers. In 2024, in Hawai'i, this is simply inaccurate. Renewable energy is an improvement over the status quo in energy, for both financial and environmental reasons.

Who pays for new renewable energy projects?

While many might believe large new renewable projects are paid for by the government or by the utility, most often they are paid for by a developer. The process is best explained by an example.

Suppose there is a company, Solar Company, which wants to build Solar Project in O'ahu. When the local utility wants to purchase more electricity, they advertise a Request for Proposal (RFP), and Solar Company submits a proposed project. In a process guided by the Public Utilities Commission (PUC), the project may be approved. Solar Company is responsible for financing and building their project – and bears the risks of a project not being completed. Solar Company and the local utility negotiate on a price to be paid for the generation and the PUC reviews and approves the terms of the agreement. PPAs are typically 20-to-25-year deals that require a specific amount of electricity to be available to the utility.

If Solar Project's total lifetime costs are \$200 Million (including financing and tax incentives) to build a 50 MW solar and storage facility that generates 2,190,000 MWh over its lifetime. Over the lifetime of the project, it would cost 9¢ per kWh for Solar Company to generate electricity. This calculation is referred to as the Levelized Cost of Energy (LCOE). After accounting for profit and risks associated with construction, Solar Company may sign a Power Purchase Agreement (PPA) to sell electricity to the utility for 11¢ per kWh. For reference, AES Kūihelani, a new 60 MW Maui-located solar and battery storage project currently has a PPA of 8.1¢ per kWh. On O'ahu, in FY2023, Kalealoa Partners, which burns low sulfur fuel oil, sold electricity at an average cost of 24¢ per kWh.

So the utility doesn't generate its own electricity?

The utility does have its own generation facilities, and it currently operates many of the older fossil fuel-based generation. The utility can also submit its own bid into the RFP process and demonstrate to the PUC that they can create a facility at a comparable cost to developers. Moreover, the utility buys, sells, and distributes energy.

Don't we just pay for renewable energy in our taxes?

The state does support new renewable energy through the Renewable Energy Technologies Income Tax Credit (RETITC) – which is available to everyone! The RETITC gives either a tax credit of 35% of the cost of a new renewable energy

system, up to \$5,000 dollars for single-family residences, or \$500,000 for commercial properties. However, the RETITC is no longer available for projects greater than 5 MW, or those requiring a PPA with the PUC. Federal tax incentives, however, are available to households, commercial entities, and developers.

Don't we just pay for renewable energy on our bill?

The only additional on-bill payment for renewable energy is used to support energy equity. There is a very small charge (\$1.18 per month) on bills which supports the State of Hawaii Green Energy Market Securitization (GEMS) program for Hawaiian Electric serviced customers only. GEMS provides low-cost loans to those who cannot afford upfront costs or cannot qualify for other financing for green infrastructure improvements. Beyond the GEMS fee, you pay for renewable energy exactly how you would pay for fossil fuels (minus the fuel costs).

Renewable Energy Generation is way more expensive than fossil fuels!

Generally, no. Although this can be a little complicated and depends on the types of renewable energy and the time it is used. Most new renewable energy projects would cost about the same as building new fossil fuels, even before accounting for subsidies. However, since fossil fuel burning power plants are already built, they can be cheaper to operate. However, in Hawai'i the price of fuel alone can be greater than the price of new renewable energy technologies. Recall that AES Kuihelani energy costs 8.1 cents per kWh.

The below graphic compares the price of operating current power plants (from Hawaiian Electric) using the range prices paid over the past 4 years with the cost of building completely new generation sources. The national levelized cost estimates are for unsubsidized renewable energy being built brand new compared with fossil fuel plants that have been in operation for decades, and the price is still comparable. However, most of the fossil fuel plants in Hawai'i have been operating well beyond their expected lifetime and will need to be retired well before the RPS goal of 2045.

Renewable Energy is unreliable and unpredictable.

Fossil fuels are unpredictable, specifically in price! Many renewable energy generators require high up-front construction and equipment costs accompanied by very low operating costs thereafter. Once you build a solar field, it keeps producing energy with only light maintenance. Fossil fuels, however, require fuel! As of 2022, Hawai'i's fossil fuel use is dependent on petroleum (oil) and oil price is very volatile. In April 2020, the spot price of oil reached as low as \$15.34 per barrel, and in March 2022, it peaked at \$140.64 per barrel. The price of your electricity bill depends directly on the price of oil.

Renewable Energy is unreliable.

We are isolated on an island, and our power is isolated onto 6 islands. If something goes down, we can't buy more power from another region like you can in the mainland. Our modern grid will be strengthened by a diversity of resources. Large-scale power plants can be cost-effective, but they simultaneously create vulnerability. This January, during a period of heavy rains, rolling blackouts occurred across the island due to the inundation of water for two of the biggest thermal (oil) generators on the island. On Hawai'i Island, a volcanic eruption causing temporary closure of Puna's geothermal plant resulted in a loss of nearly 30% of the grid's generation capacity. But the growth of household solar, multiple solar and wind farms, biofuel and biomass-based generators, and battery storage reduces vulnerability and generates resiliency.

Figures from the original document

Figures are reproduced without substantive alteration.

